PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The impact of modifiable reproductive factors on cancer incidence and mortality in Korea: A systematic review protocol
AUTHORS	Kim, Seo-Hee; Han, Mi Ah

VERSION 1 – REVIEW

REVIEWER	Butt, Salma Lunds Universitet, Dept of Surgery
REVIEW RETURNED	19-Sep-2022

GENERAL COMMENTS	No study has yet been performed. Interesting topic and sould definetly be studied. One comment, age at first birth has increased not decreased. Will be happy to review study when completed.

REVIEWER	Manouchehri, Elham Mashhad University of Medical Sciences
REVIEW RETURNED	27-Oct-2022

GENERAL COMMENTS	While wishing success to respected authors, comments are provided in the attached file – contact publisher to view this file.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Salma Butt, Lunds Universitet

Comments to the Author:

No study has yet been performed. Interesting topic and sould definetly be studied. One comment, age at first birth has increased not decreased. Will be happy to review study when completed.

RESPONSE: We revised the sentence in the Abstract as follows:

> *The features of reproductive factors have changed, which include increased age at first childbirth and decreased breastfeeding duration.*

Reviewer: 2

Dr. Elham Manouchehri, Mashhad University of Medical Sciences, Mashhad University of Medical Sciences

Comments to the Author:

While wishing success to respected authors, comments are provided in the attached file.

Abstract

According to available studies, decreasing age at first childbirth is not associated with an increased risk of cancer. This article is also mentioned in the introduction, which contradicts this sentence.

RESPONSE: We revised the sentence in the Abstract as follows:

The features of reproductive factors have changed, which include increased age at first childbirth and decreased breastfeeding duration.

Search databases web of science and Scopus are also recommended.

RESPONSE: We have already completed the data search and added associated text as follows:

We conducted a literature search in the EMBASE, MEDLINE, Cochrane Library, and Korean databases such as the Korean Studies Information Service System, Research Information Sharing Service, KoreaMED, Korean Medical Database, National Assembly Library, and Korea Institute from their inception to August 24, 2022.

Introduction

This does not apply to all types of cancer. for example older age at first birth is associated with a lower incidence of breast cancer.

RESPONSE: We appreciate the reviewer's comments. And we added associated text as follows:

However, different associations have been observed for different types of cancer, such as a lower risk of breast cancer with increasing age at first birth.[12]

Methods

What is meant by childbirth?
If you mean the number of births mentioned below. If you mean to be nulliparous or parous, please use a better word for example having childbirth or nulliparous/parous.

RESPONSE: We meant the childbirth history and revised the text as follows:

(3) studies considering at least one of the following modifiable reproductive factors as an exposure: childbirth history (nulliparous or parous), age at first childbirth, age at last childbirth, number of childbirths, breastfeeding experience, and duration of breastfeeding;

It is suggested to include elective abortion and elective cesarean section in the included reproductive factors.

RESPONSE: We appreciate the reviewers' comments. However, studies might not have differentiated between elective and emergency abortions and CS, and even elective abortions and CS might be for medical treatment purposes. Therefore, we decided to not include them.

Please specify the time range and desired languages.

reference lists of included studies can be searched manually.

RESPONSE: According to the reviewer's comments, we added associated method as follows:

We will include publications in Korean and English, and manually search the reference lists of included studies.

Please specify the study selection process will be illustrated by means of a PRISMA flowchart.

RESPONSE: According to the reviewer's comments, we added associated method as follows:

We will use a PRISMA flow diagram to illustrate the study selection process.

VERSION 2 – REVIEW

REVIEWER	Manouchehri, Elham Mashhad University of Medical Sciences
REVIEW RETURNED	15-Nov-2022

GENERAL COMMENTS	While wishing success to the respected authors of the protocol and the theoretical accuracy they did in answering the suggestions point by point, only one item needs to be corrected. In this sentence: "However, different associations have been observed for different types of cancer, such as a lower risk of breast cancer with increasing age at first birth.12" please replace "lower" with "higher". Regarding the impact of abortion on the incidence of cancer, spontaneous abortions are reported separately from induced abortions in the literature. If there is this condition in the included articles, it is suggested to investigate the effect of induced abortion on cancer.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. Elham Manouchehri, Mashhad University of Medical Sciences, Mashhad University of Medical Sciences

Comments to the Author:

While wishing success to the respected authors of the protocol and the theoretical accuracy they did in answering the suggestions point by point, only one item needs to be corrected.

In this sentence: "However, different associations have been observed for different types of cancer, such as a lower risk of breast cancer with increasing age at first birth.12" please replace "lower" with "higher".

RESPONSE: The sentence was revised as the reviewer commented.

Regarding the impact of abortion on the incidence of cancer, spontaneous abortions are reported separately from induced abortions in the literature. If there is this condition in the included articles, it is suggested to investigate the effect of induced abortion on cancer.

Best wishes

Response: We appreciate the reviewer's comments. Although studies may have distinguished between spontaneous and induced abortions, induced abortions may have been performed for therapeutic purposes. And, as mentioned in the first revision, the search has been completed.